

Divisibility, GCD and LCM

Diagnostic and coverage-hardening companion

TARGET → DIVISOR / MULTIPLE → REDUCTION → THEOREM CHECK → VERIFY

Teacher-only companion - answer diagnostics and theorem-hypothesis checks

NT-01 - Teacher Diagnostic Key

This key is teacher/authoring material. It is not included in the student PDF.

Recognition and First-Line Lab

1. $d \mid (296 - 188)$, so $d \mid 108$.
2. $N = \text{lcm}(14, 20, 35)$ for the least positive common multiple; value 140 if completed.
3. $12345 = 2(5432) + 1481$; continue Euclid.
4. $ab = 9 \cdot 252 = 2268$.
5. $3(5x + 7y) - 7(2x + 3y) = x$, so $d \mid x$. Also $5(2x + 3y) - 2(5x + 7y) = y$.
6. $N - 4 = \text{lcm}(6, 10, 15) = 30$ for the least solution, so $N = 34$ if completed.
7. $d \mid (1046 - 812) = 234$ and $d \mid (1280 - 1046) = 234$.
8. $\text{lcm}(18, 24) = 72$ minutes.
9. $\text{gcd}(u, v) = 1$.
10. $1260 = 84 \cdot 15$.
11. $7(4n + 7) - 4(7n + 13) = -3$, so $d \mid 3$.
12. $6 \mid x \mid 72$.
13. Differences are $1457 - 1001 = 456$ and $1913 - 1457 = 456$; start with their gcd.
14. $ab = 12 \cdot 420 = 5040$; stop because only the product is requested.
15. $a = 12u$, $b = 12v$, $\text{gcd}(u, v) = 1$, $uv = 420/12 = 35$.
16. Write one division-with-remainder equation $a = qb + r$ and replace the gcd pair by (b, r) .

Recognition diagnostic

- NT01-R1: divisor-vs-multiple confusion.
- NT01-R2: same-remainder fork not recognized.
- NT01-R3: Euclidean reduction not recognized as invariant-preserving.
- NT01-R4: structural divisibility replaced by an irrelevant digit test.
- NT01-R5: gcd/lcm product used without deciding whether reconstruction is required.
- NT01-R6: divisibility chain/transitivity missed.

Practice and Transfer Bank

F0

1. Yes: $234 = 18 \cdot 13$.
2. $48 - 30 = 18$; common divisibility is closed under subtraction.
3. $\text{gcd}(84, 126) = 42$, $\text{lcm}(84, 126) = 252$.
4. From $8 \mid 24$ and $24 \mid 120$, $8 \mid 120$. Also $\text{gcd}(24, 120) = 24$ and $\text{lcm}(24, 120) = 120$.

F1

5. $239 - 155 = 84$; greatest divisor 84.
6. $\text{lcm}(16, 20, 24) = 240$.
7. Euclid gives $\text{gcd}(2025, 748) = 1$.
8. $\text{lcm}(12, 18, 30) = 180$; $N = 7 + 180 = 187$.
9. $ab = 12 \cdot 420 = 5040$.

F2

10. Differences are 144 and 144; answer 144.
11. Differences are 168 and 168. All positive divisors of 168 work: 1, 2, 3, 4, 6, 7, 8, 12, 14, 21, 24, 28, 42, 56, 84, 168.
12. $\text{lcm}(12, 15, 20) = 60$; $N = 69$.
13. Normalize: $a = 12u$, $b = 12v$, $\text{gcd}(u, v) = 1$, $uv = 35$. Unordered pairs: $(12, 420)$ and $(60, 84)$.
14. Write $x = 6k$ and require $6k \mid 72$, hence $k \mid 12$. Values: x in $\{6, 12, 18, 24, 36, 72\}$.

F3

15. From $2(7x + 5y) - 5(3x + 2y) = -x$ and $3(7x + 5y) - 7(3x + 2y) = y$, any common divisor divides both x and y . If $\gcd(x, y) = 1$, only $d = 1$ is possible; therefore no $d > 1$ satisfies the premise.
16. $\text{lcm}(18, 24, 40) = 360$ minutes; next reset is 15:00.
17. $7(4n+7) - 4(7n+13) = -3$, so $d \mid 3$. Values 1 and 3 are both attainable; for $d = 3$, take $n \equiv 2 \pmod{3}$ in ordinary remainder language.
18. $uv=900/15=60$, $\gcd(u, v) = 1$. Unordered coprime splits: $(1, 60), (3, 20), (4, 15), (5, 12)$. Thus $(a, b) = (15, 900), (45, 300), (60, 225), (75, 180)$ up to order.
19. Differences are 456 and 456; largest step 456 mm.

F4

20. 456.
21. $N+5$ must be a multiple of $\text{lcm}(18, 24, 30) = 360$; least positive $N = 355$.
22. $11(7n + 5) - 7(11n + 8) = -1$; largest possible positive divisor is 1.
23. Let $x = \gcd(a, c)$, $y = \gcd(b, c)$. The relation reduces to $a(27/x-26)+b(27/y-26)=0$. Exactly one of x, y is 1. Under $a, b \leq 50$, the other is forced to 2, yielding the two symmetric core pairs $(25, 2)$ and $(2, 25)$ before counting admissible c . A completed independent count gives 40, matching the validated anchor.
24. $N-5$ is a multiple of $\text{lcm}(12, 18) = 36$, so $N = 5 + 36k$. Requiring $5 \mid N$ gives k divisible by 5. For $N > 5$, least $k = 5$, hence $N = 185$.

Transfer prompts

25. $d \mid (a - b)$.
26. Largest equal spacing is a common-divisor target $\rightarrow \gcd$; first synchronization is a common-multiple target $\rightarrow \text{lcm}$.
27. Product is fixed: $ab = 18 \cdot 630 = 11340$. The actual pair is not unique because after $a = 18u, b = 18v$, one has $uv = 35$, $\gcd(u, v) = 1$, allowing $(u, v) = (1, 35)$ or $(5, 7)$ and their reversals.
28. Retrieve $d \mid (a - b)$ from the equal-remainder condition; do not reteach modular legality here.
29. The target is a divisor of differences, not a common multiple. Start with differences 144, 144 and $\gcd$ them.
30. $gL = ab$ alone does not determine a unique pair. Normalize and enforce $\gcd(u, v) = 1$; $(60, 84)$ is a second unordered pair besides $(12, 420)$.

H0 Mastery Test

1. $d \mid (596 - 428) = 168$ and $d \mid (764 - 596) = 168$.
2. $N - 11 = \text{lcm}(18, 24, 30)$.
3. $8(13n + 9) - 13(8n + 5) = 7$, so $d \mid 7$.
4. $ab = 14 \cdot 840 = 11760$.
5. Euclid gives $\gcd(123456, 7890) = 6$.
6. 168.
7. $\text{lcm}(18, 24, 30) = 360$; $N = 371$.
8. 1760.
9. $11(7n + 5) - 7(11n + 8) = -1$; answer 1.
10. All positive divisors of 168: 1, 2, 3, 4, 6, 7, 8, 12, 14, 21, 24, 28, 42, 56, 84, 168.
11. $ab = 12 \cdot 360 = 4320$; with $a = 72$, the other number is 60. Check: $\gcd(72, 60) = 12$, $\text{lcm}(72, 60) = 360$.

12. $lcm(28, 42, 63) = 252$ days.
13. A: differences are 102 and 102 $\rightarrow$ greatest divisor 102. B: $N - 5 = lcm(6, 10, 15) = 30 \rightarrow$ least $N = 35$. First moves differ because A asks for an unknown divisor, while B constructs an unknown number with a prescribed remainder.
14. A: digit sum of 7425 is 18, so it is divisible by 9. B: the linear combination is $7(4n+7) - 4(7n+13) = -3$, so possible positive d are 1 and 3, both attainable. A is a local one-number test; B is a structural common-divisor relation.
15. Differences are 234 and 234, so the largest unit is 234 m. Hidden model: equal offsets mean equal remainders, so the unit divides all position differences.
16. Counterexample: item 6 mentions same remainders but asks for the divisor, so the correct route is gcd of differences. Decision boundary: identify whether the unknown is the divisor or the number being constructed.

Source-anchor audit

IOQM-2025-Q02

Independent count: $\text{floor}(100/3) - \text{floor}(100/6) = 33 - 16 = 17$. Final official key: 17. PASS.

IOQM-2025-Q27

Independent reduction through $lcm(x, c) = xc / gcd(x, c)$ forces the two symmetric core pairs (25, 2) and (2, 25) and 20 admissible c values for each. Total 40. Final official key: 40. PASS.

H0 diagnostic rule

Static diagnostic only, not psychometric calibration:

- if a learner repeatedly needs a method label before writing a first useful line, route back to the First-Step Reference;
- if the learner calculates correctly but confuses divisor/multiple or unknown-divisor/unknown-number boundaries, classify as a recognition error rather than an arithmetic error;
- no classroom time threshold, pass probability, percentile or psychometric cutoff is claimed here.

NT-01 - Teacher Coverage Hardening Addendum

Status: STATIC_DIAGNOSTIC_ADDENDUM_V1 Issue: #132

This addendum synchronizes the learner-facing Euclid's Lemma bridge added to 03_First_Step_Reference.md. It does not alter historical-anchor answers or existing authored-item keys.

Euclid's Lemma

Statement:

If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Independent proof check

If p does not divide a , primality implies $gcd(p, a) = 1$. By Bezout/linear-combination structure there are integers r, s with

$$rp + sa = 1.$$

Multiplying by b gives

$$rpb + sab = b.$$

Since p divides both left-hand terms, $p \mid b$. Thus either $p \mid a$ or $p \mid b$.

Required diagnostic contrast

The primality hypothesis is necessary. $6 \mid 2 \cdot 3$, but 6 divides neither factor. A learner who applies the lemma to an arbitrary composite divisor has not checked the theorem hypothesis.

Diagnostic codes

- NT01-EUCLID-1: prime hypothesis not checked.
- NT01-EUCLID-2: composite divisor incorrectly split across a product.
- NT01-EUCLID-3: theorem name recalled but no link to divisibility/gcd structure.
- NT01-EUCLID-4: Euclid's Lemma confused with the Euclidean algorithm.

Downstream boundary

NT-03 may retrieve Euclid's Lemma as an NT-01 export when prime divisibility must split across a product. NT-03 remains the canonical owner of prime factorisation, valuations, divisor counts and perfect-power structure.

Evidence truth

This addendum is a static mathematics/diagnostic synchronization. It does not constitute classroom timing/readability, retention, psychometric, qualification/pass-mark or publication calibration; those remain NOT_RUN.